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Automotive Motor Control



The reference design combines a microcontroller with analogue components in a compact package. With power management, interfaces, and debugging capabilities, it offers a solution for efficient, high-performance applications.

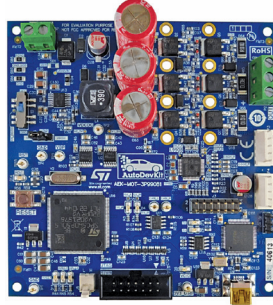
The S32M276SFRRD is a reference design from NXP Semiconductors that incorporates the S32M276 integrated system, which combines a 32-bit ARM Cortex-M7 S32K3 microcontroller and an analogue component, including a voltage regulator, gate driver, and current sensor, within a single system-in-package design. The design includes interfaces such as UART and CAN with an internal PHY, RGB LED, JTAG 10-pin connector, faston connectors for power supply, and a 3-phase motor connection, and it operates with an external +12V/5A supply. The internal analogue extension handles the power management die within the SiP, distributing the power from the 12V supply, minimising BOM and conserving space. For MCU programming, it features JTAG connectors compatible with a PE Multilink universal FX debug probe, allowing users to control the target's execution, debug code, and program memory. The microcontroller integrates a digital programmable gain amplifier for low-side current measurements, which includes a current sense amplifier, adjustable gain, blanking time counter, and over-current comparator.

The reference design is suitable for automotive BLDC and PMSM motor control applications.

NXP Semiconductors

<https://www.electronicshobby.com/electronics-projects/reference-design-for-automotive-motor-control>

Low Voltage BLDC Automotive Motor Control



The reference design integrates motor speed detection, field-oriented control, and communication via CAN, offering a solution for motor management while ensuring safety and flexibility in operation.

The AEK-MOT-3P99081 evaluation board from STMicroelectronics supports encoder inputs and Hall sensors for motor speed detection and control. The drive system uses a field-oriented control (FOC) algorithm for operation. Controlled via a standard CAN command set from any domain or zone controller, the design simplifies the management of control loop bandwidth and resource demands. The CAN protocol enables data exchange, allowing commands to start or stop the motor, adjust its speed, reverse direction, and relay error notifications. Programming is possible via an integrated programmer/debugger or an external adaptor connected to the JTAG connector. The board includes a gate driver and a three-phase motor gate driver, with features for BLDC motor control, including current monitors for ground-referred currents. It also ensures safety with a power supply unit that protects against battery polarity inversion and power generation through a DC-DC converter and voltage regulators.

The reference design is suitable for brushless motors in automotive applications.

STMicroelectronics

<https://www.electronicshobby.com/electronics-projects/low-voltage-bldc-automotive-motor-control-reference-design>

Compact Automotive BLDC Motor



The motor controller reference design handles power, controls torque and speed, and includes safety features, with communication options and support for motor control techniques like FOC.

The dsPIC33CK64MP105 DSC motor controller reference design by Microchip Technology provides a platform for engineers to develop motor control systems with continuous 50A RMS and peak current up to 150A. It supports an 18V–85V input voltage range, suitable for up to a 20s battery pack, and features an XT90-type connector. Based on field oriented control (FOC) with Hall sensors, the system ensures precise motor torque and speed control. Integrated features, such as maximum torque per amp (MTPA) for IPMSM motors and field weakening (FW), extend the power range. The current sensing mechanism employs three 500μΩ low-side shunt resistors per phase and configurable PWM switching frequencies from 8kHz to 20kHz. Safety protections include overvoltage and undervoltage safeguards, motor phase overcurrent detection, and over-temperature protection on the inverter stage MOSFETs. The controller consists of CAN communication and auxiliary connections for throttle input, voltage monitoring, I2C, UART, and Bluetooth options for connectivity.

The reference design is suitable for micro-mobility applications where space is at a premium.

Microchip

<https://www.electronicshobby.com/electronics-projects/compact-bldc-motor-reference-design>